

CS1CA Computer Systems Architecture
Storage systems and networking (2)
Week 12 lecture 2 of autumn semester 2025

Lectured by

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Outline of contents in the topic

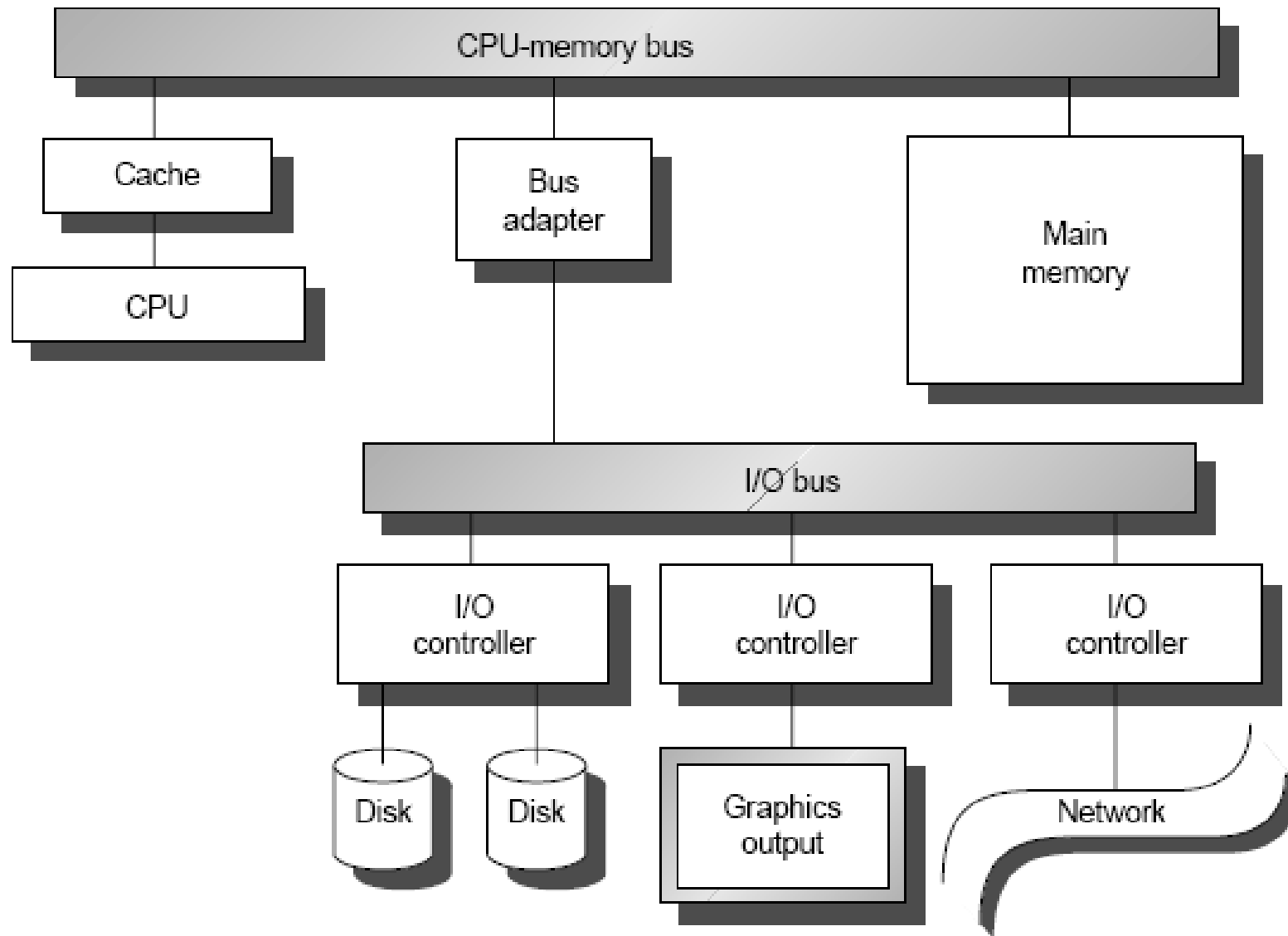
- Magnetic disk (Solid State Drive, SSD, is not covered in the topic, but you are encouraged to learn it by yourself)
 - Structure and management
 - Disk performance measure
- Buses
- I/O and I/O performance measures
- Reliability, availability, and RAID
- A 2-node simple network: performance of the simple network

In this lecture

- The following contents will be delivered
 - I/O and I/O performance measures
 - Reliability, availability, and RAID
 - A 2-node simple network: performance of the simple network
- Expected Learning outcomes are listed below.
 - To gain knowledge of structure and organization of magnetic disks, and to be able to measure disk performance
 - To have the knowledge of bus design options

Interfacing storage devices to the CPU

- Question: how should the physical connection of the I/O bus be made?
- Two choices
 - I/O bus connecting to memory;
 - I/O bus connecting to cache.
- Two addressing schemes for the CPU to address I/O devices.
 - Memory-mapped I/O;
 - Dedicated I/O opcodes in the CPU.



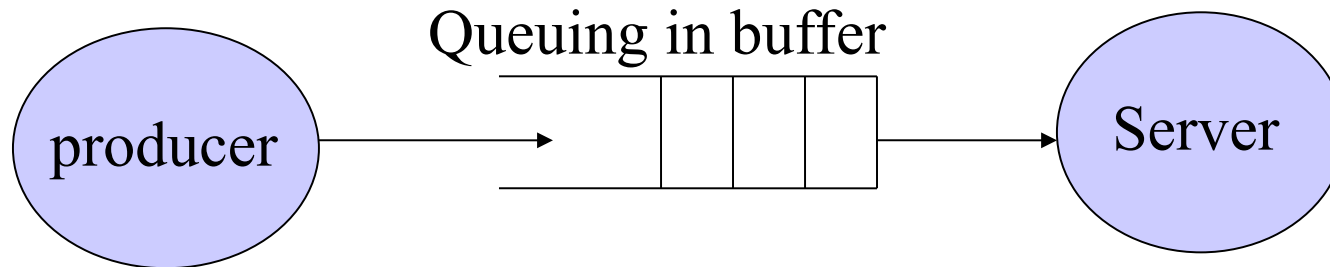
A typical interface of I/O devices and an I/O bus to the CPU memory bus

Direct memory access (DMA) hardware

- DMA is a processor that transfers data between memory and an I/O device while the CPU goes on with other tasks.
- CPU sets DMA registers:
 - Memory address;
 - Number of bytes to be transferred.
- DMA becomes a bus master; it interrupts CPU after the transfer is completed;
- There may be several DMA devices.

I/O performance measures

- **The simple producer-server model**



- Producer creates tasks to be performed and places them in a buffer;
- The server takes tasks from the FIFO buffer and executes them.
- **Response time**: the time between a task is placed in the buffer and the server finishes the task;
- **Throughput**: average number of tasks completed by the server over a time period.

A Little queuing theory

- Assumption: the input rate must be equal to the output rate in a long-term state of a system.
- **Little's Law**

$$\text{Mean number of tasks in system} = \text{Arrival rate} \times \text{Mean response time}$$
$$\text{Server utilisation} = \text{Arrival rate} / \text{service rate}$$

Example

- An I/O system with a single disk gets about 10 I/O requests per second;
- The average time for a disk to service an I/O request is 50 ms;
- What is the utilisation of the I/O system?

Service rate = $1/50\text{ms} = 20$ I/O per second

Server utilisation = *arrival rate / service rate*
 $= 10/20 = 0.5$

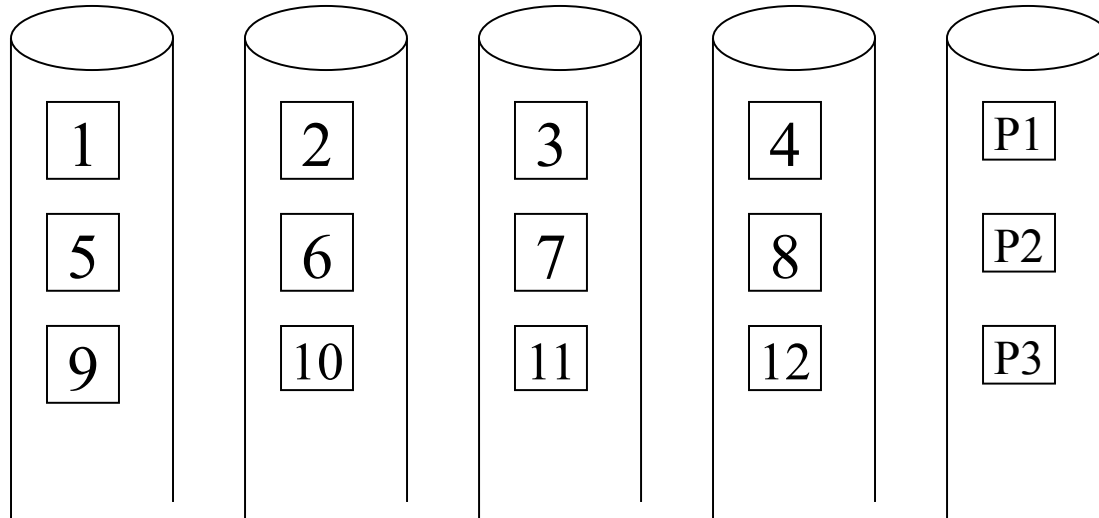
Storage Systems: Reliability, Availability, and RAID

- **Reliability:** is anything broken?
 - Reliability can only be improved by building better components.
- **Availability:** is system still available to user?
 - Adding hardware can improve availability.
- **RAID:** Redundant Array of Independent Disks is a set of physical disk drives viewed by the operating system as a single logical drive.

RAID: Solutions to disk failure

- **Mirroring:** copy information to another redundant disk. It is an expensive method.
- **Block-interleaved parity:**
 - Rather than a complete copy of all disks, use only enough redundant information to restore the lost information on a failure;
 - One disk with multiple blocks serves as the redundant disk for all other disks;
 - Parity scheme: the missing information can be reconstructed by checking the ‘parity’.

Example: block-interleaved parity



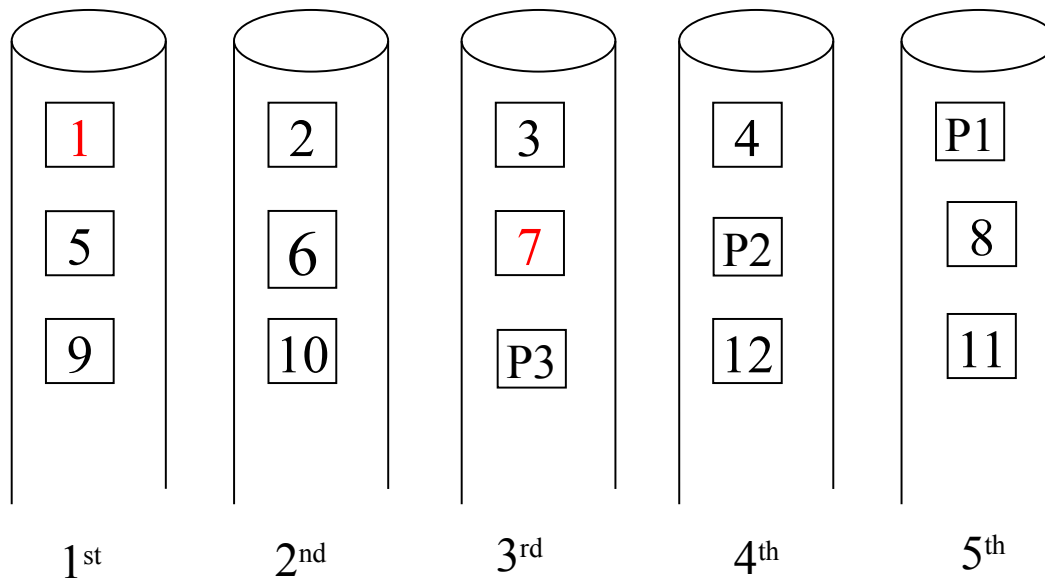
- Block 1 has information 0011;
- Block 2 has information 0110;
- Block 3 has information 0100;
- Block 4 has information 0001.

Sum of each bit starting from the least significant bit.

Parity P1 is the sum MOD 2 (bit-wise operation), as 0000.

- **Block-interleaved distributed parity**

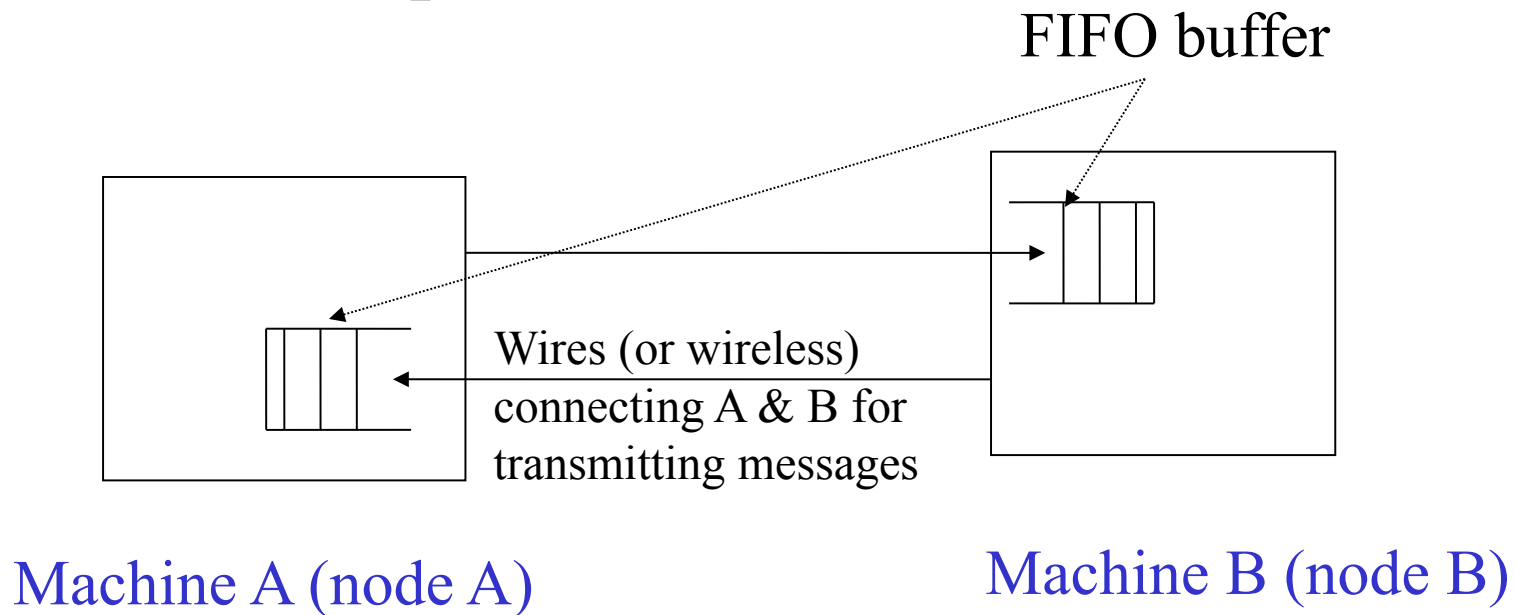
- The difference is that independent access can occur in parallel: the redundant information is distributed into all disks as redundant blocks;
- Multiple writes can occur simultaneously.



- Writing to block 1 accesses 1st and 5th disks;
- Writing to block 7 accesses 3rd and 4th disks;

Interconnection networks

- A 2-node simple network



Performance of the simple network

- Performance of the simple network is measured by total latency, which is defined as

$$\textit{Total latency} = \textit{sender overhead} + \textit{time of flight} + \textit{transmission time} + \textit{receiver overhead}$$

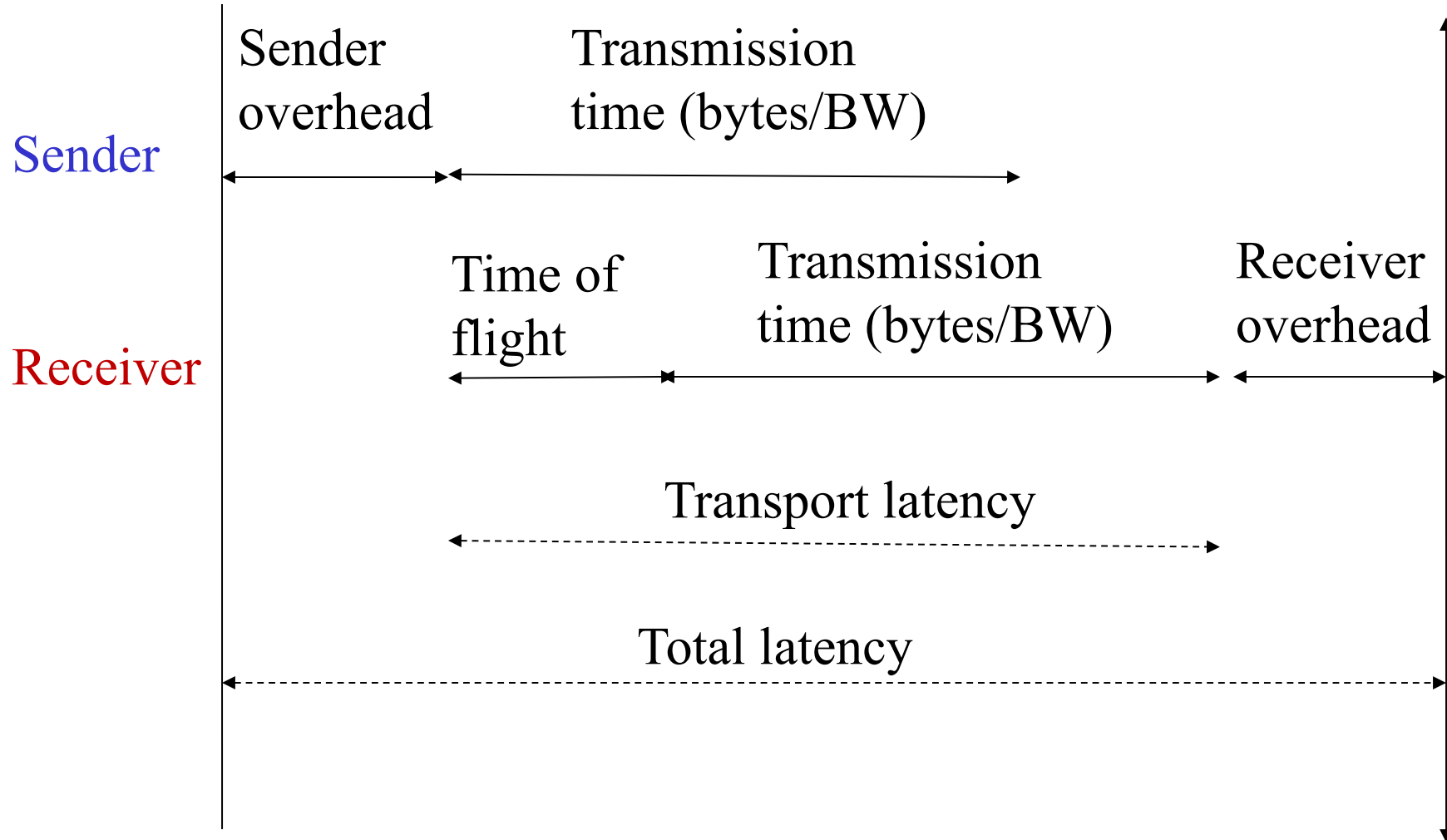
where

- *Sender overhead*: time for the processor to inject the message into the interconnection network;
- *Time of flight*: time for the first bit of message to arrive at the receiver end, which is
$$= \textit{distance between two nodes} / \textit{propagating speed}$$

(Continue.....)

- Transmission time: time for the message to pass through the network, which is
$$= \text{size of message} / \text{bandwidth}$$
- Bandwidth: maximum rate at which the interconnection network can propagate information in megabits per second;
- Receiver overhead: time for the processor to pull the message from the interconnection network;
- Transport latency: time the message spends in the interconnection network;

Performance parameters of interconnection networks



Example:

- Two machines connected together;
- Network bandwidth = 10 Mbits/sec.
- Sender overhead = 230 μ s
- Receiver overhead = 270 μ s
- A single message of 1000 bytes is to be sent from one machine to the other.
- Signals propagate at about 50% of the light speed (3×10^8 m/sec) in a conductor

Calculate the latency when the two machines are 100 meters apart. How about 1000km apart?

Answer:

Old network saying

Bandwidth problems can be cured with money; Latency problems are harder because the speed of light is fixed – you can't bribe God.

David Clark, MIT

Effective bandwidth to guide a design

- Simplify the performance equation by combining sender overhead, receiver overhead, and time of flight into a single term called *Overhead*:

$$\text{Total latency} = \text{Overhead} + \frac{\text{Message size}}{\text{Bandwidth}}$$

- The effective bandwidth delivered by the network as message size varies is defined as:

$$\text{Effective bandwidth} = \frac{\text{Message size}}{\text{Total latency}}$$

End of this lecture

Questions and exercises

- What is DMA used for?
- How to measure I/O performance? (Little's law)
- What is RAID, and what does it do?
- In a 2-node simple computer network, how to measure its performance?